INTERNATIONAL MONETARY FUND

HAITI

April 13, 2020

REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY—DEBT SUSTAINABILIT ANALYSIS

Approved By
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Prepared by the staffs of the International Monetary Fund and the International Development Association.

Haiti: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	High
Overall risk of debt distress	High
Granularity in the risk rating	Debt is sustainable in the medium-term. Long-term sustainability will require fiscal consolidation, including through higher domestic revenue mobilization.
Application of judgment	Yes: High probability of protracted and substantial threshold breaches from FY2033.
Macroeconomic projections	Compared to the December 2019 DSA, growth in the short-term has been revised down and inflation revised up. The projected fiscal deficit as a percent of GDP is 3.0, 0.8 and 0.9 percent higher respectively in FY2020, 2021-24, and 2025-35. The projected current account deficit is higher in the short term, but 0.5 percent of GDP smaller on average over 2025-35.
Financing strategy	Future gross financing needs are assumed to be met both internally—by the rollover of central bank advances to the government—and externally. Remaining internal financing takes the form of short-term treasury bills held by commercial banks. External debt financing, contracted or guaranteed, is assumed to be mostly concessional.
Realism tools flagged	The baseline assumptions are credible, and the projected fiscal adjustment is realistic.
Mechanical risk rating under the external DSA	Moderate risk
Mechanical risk rating under the public DSA	Moderate risk

The Debt Sustainability Analysis (DSA) was prepared in accordance with the revised joint Bank-Fund debt sustainability framework (DSF) for low-income countries (LICs).[1] It updates the DSA prepared for the 2019 Article IV Consultation.[2]

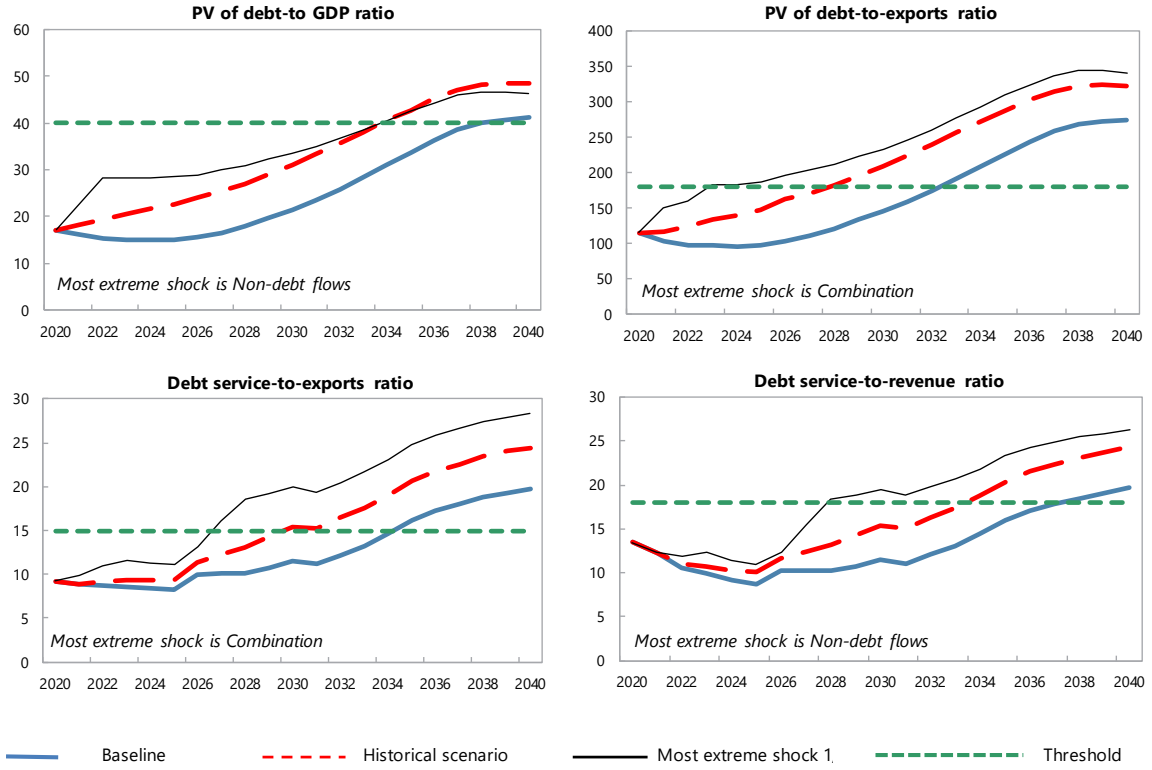
Haiti's risk of debt distress is assessed to be "high", although the model-based risk rating for both external and overall public debt is "moderate." An application of judgement was applied to change the rating from "moderate" to "high" because of the high probability of threshold breaches under the baseline scenario from FY2033, and by Haiti's institutional fragilities and exceptional vulnerability to natural disasters. Haiti is an FCV country—a country affected by fragility, conflict, and violence as defined by the World Bank—and tailored stress tests suggest that its debt risk rating is very vulnerable to large natural disaster shocks which are statistically very frequent.

Nevertheless, the moderate level of public debt and broadly stable debt trajectory over the next ten years point to sustainable public debt.

[1] Guidance Note on the Bank-Fund Debt Sustainability Framework for Low-Income Countries, February 2018.

[2] This DSA assumes that debt service to the IMF falling due in the 24 months from April 14, 2020 will be covered under the Catastrophe Containment and Relief Trust Fund (CCRTF), subject to availability of resources and decisions of the Executive Board of the IMF.

Figure 1. Haiti: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2020–40



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	Yes	No
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.7%	1.7%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	23	23
Avg. grace period	5	5

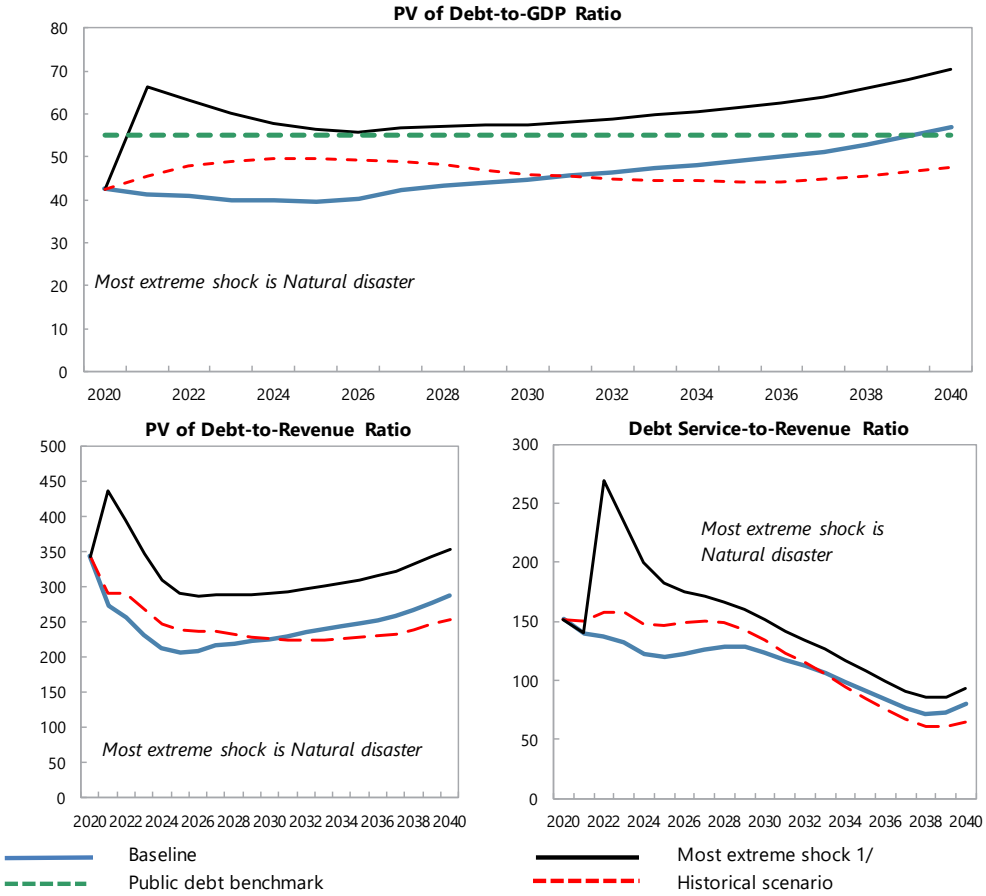
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most extreme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Haiti: Indicators of Public Debt Under Alternatives Scenarios, 2020–40



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	8%	8%
Domestic medium and long-term	0%	0%
Domestic short-term	92%	92%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.7%	1.7%
Avg. maturity (incl. grace period)	23	23
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.0%	0.0%
Avg. maturity (incl. grace period)	1	1
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	-11.0%	-11.0%

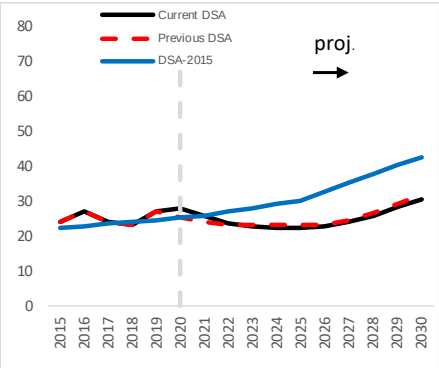
* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

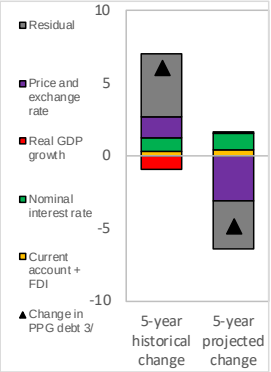
1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most extreme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 3. Haiti: Drivers of Debt Dynamics-Baseline Scenario External Debt

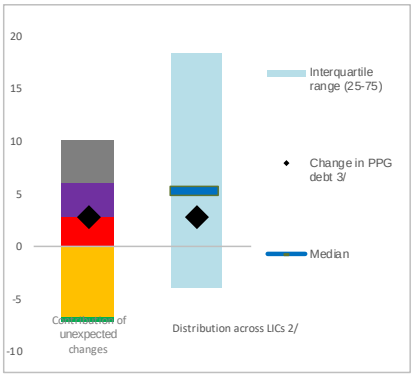
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

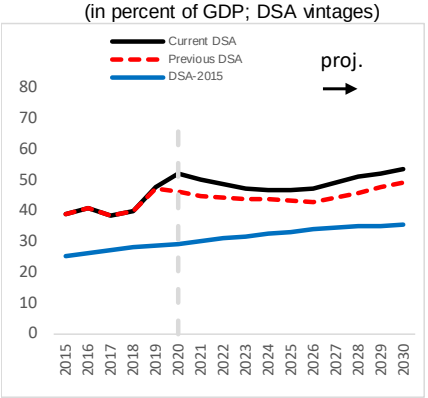


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

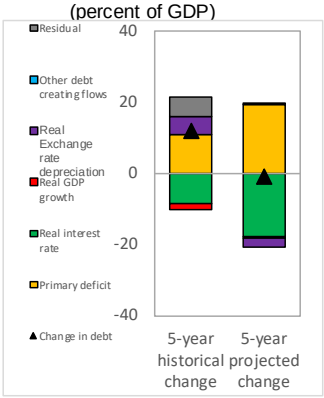


Public debt

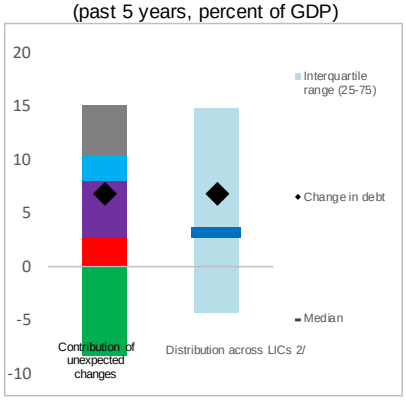
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



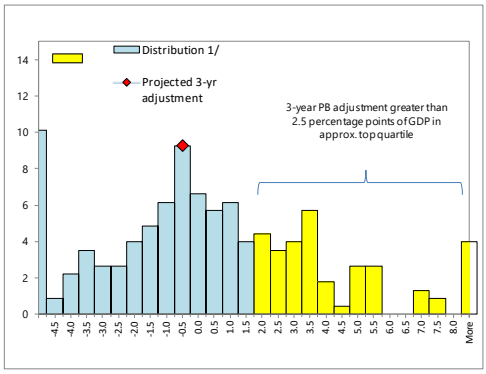
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

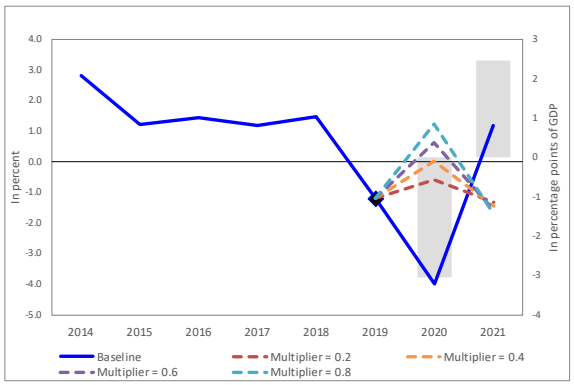
Figure 4. Haiti: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



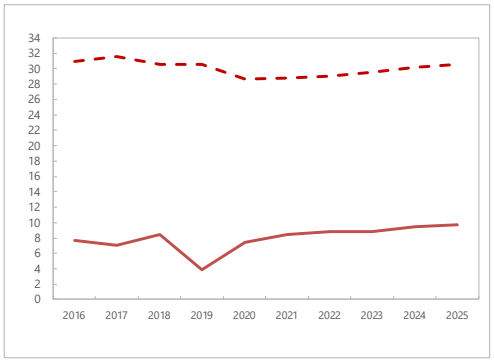
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

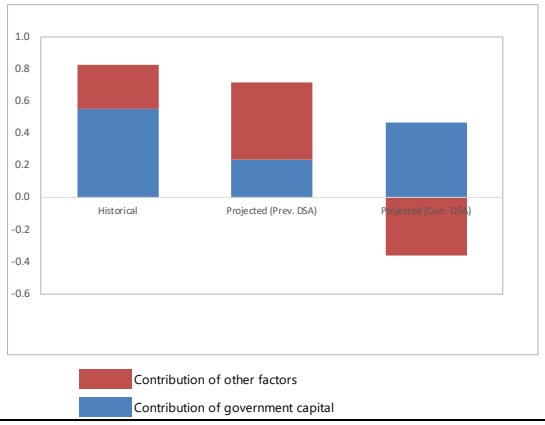
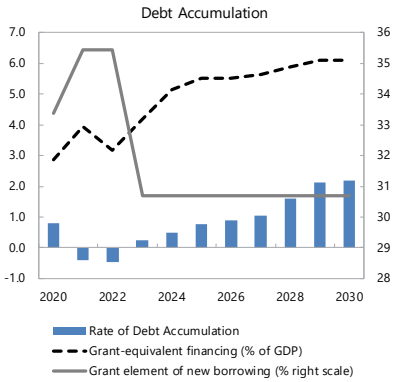


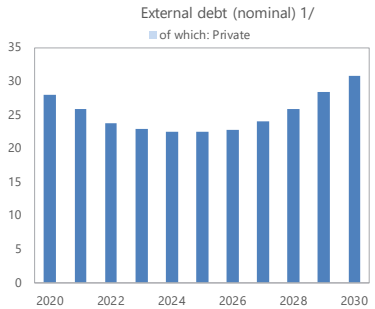
Table 1. Haiti: External Debt Sustainability Framework, Baseline Scenario, 2017–40

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	**24.2**	**23.5**	**27.4**	**28.0**	**25.8**	**23.8**	**22.9**	**22.5**	**22.4**	**30.8**	**55.8**	**20.1**	**25.2**
of which: public and publicly guaranteed (PPG)	**24.2**	**23.5**	**27.4**	**28.0**	**25.8**	**23.8**	**22.9**	**22.5**	**22.4**	**30.8**	**55.8**	**20.1**	**25.2**
Change in external debt	-3.2	-0.7	3.9	0.6	-2.2	-2.0	-0.9	-0.4	-0.1	2.4	0.2		
Identified net debt-creating flows	**-4.9**	**-0.4**	**3.1**	**4.3**	**-1.2**	**-1.3**	**-0.4**	**0.2**	**0.8**	**0.6**	**0.8**	**1.6**	**0.5**
Non-interest current account deficit	**0.8**	**3.7**	**1.2**	**3.4**	**0.3**	**0.2**	**1.2**	**1.8**	**2.3**	**2.4**	**2.3**	**3.6**	**2.0**
Deficit in balance of goods and services	36.0	40.3	41.5	36.9	36.5	36.8	36.4	35.9	35.5	36.0	35.8	**37.8**	**36.2**
Exports	19.8	18.4	18.2	14.8	15.7	15.6	15.5	15.4	15.4	14.9	15.1		
Imports	55.8	58.7	59.7	51.8	52.2	52.3	51.8	51.4	50.8	50.8	50.8		
Net current transfers (negative = inflow)	-34.3	-35.9	-39.5	-33.0	-35.3	-35.7	-34.4	-33.3	-32.3	-32.5	-31.8	**-33.6**	**-33.4**
of which: official	-4.6	-4.0	-2.1	-2.2	-3.8	-3.0	-3.8	-4.6	-4.9	-4.8	-4.6		
Other current account flows (negative = net inflow)	-0.9	-0.7	-0.8	-0.5	-0.8	-0.8	-0.8	-0.8	-0.8	-1.0	-1.6	**-0.7**	**-0.8**
Net FDI (negative = inflow)	**-4.5**	**-1.1**	**-0.9**	**-0.5**	**-1.5**	**-1.5**	**-1.5**	**-1.5**	**-1.5**	**-1.8**	**-1.8**	**-1.8**	**-1.6**
Endogenous debt dynamics 2/	**-1.2**	**-2.9**	**2.8**	**1.3**	**-0.1**	**0.0**	**0.0**	**-0.1**	**-0.1**	**0.0**	**0.3**		
Contribution from nominal interest rate	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.4	1.0		
Contribution from real GDP growth	-0.3	-0.3	0.3	1.1	-0.3	-0.2	-0.3	-0.3	-0.3	-0.4	-0.8		
Contribution from price and exchange rate changes	-1.1	-2.8	2.3	...	...	...	...	...	...	...	...		
Residual 3/	**1.7**	**-0.3**	**0.8**	**-3.6**	**-0.9**	**-0.7**	**-0.5**	**-0.6**	**-0.8**	**1.8**	**-0.6**	**-0.7**	**-0.2**
of which: exceptional financing	-1.7	-0.9	-1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**16.1**	**17.1**	**16.2**	**15.2**	**14.9**	**14.9**	**15.1**	**21.6**	**41.1**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**88.6**	**115.5**	**103.5**	**97.8**	**96.6**	**96.4**	**98.3**	**145.1**	**273.3**		
PPG debt service-to-exports ratio	**5.7**	**6.1**	**7.3**	**9.2**	**8.9**	**8.8**	**8.6**	**8.4**	**8.2**	**11.6**	**19.8**		
PPG debt service-to-revenue ratio	**8.0**	**8.6**	**12.3**	**13.5**	**12.3**	**10.6**	**9.9**	**9.2**	**8.8**	**11.5**	**19.6**		
Gross external financing need (Million of U.S. dollars)	-210.9	356.4	142.3	368.7	22.8	8.4	95.6	153.2	215.3	253.7	462.6		
Key macroeconomic assumptions													
Real GDP growth (in percent)	1.2	1.5	-1.2	-4.0	1.2	1.0	1.1	1.2	1.4	1.4	1.4	**1.4**	**0.8**
GDP deflator in US dollar terms (change in percent)	4.2	13.2	-8.7	2.9	2.0	2.6	2.5	2.4	2.4	0.4	0.4	**1.7**	**1.4**
Effective interest rate (percent) 4/	0.7	1.0	0.9	0.8	0.9	0.9	0.9	1.0	1.1	1.6	1.9	**0.5**	**1.1**
Growth of exports of G&S (US dollar terms, in percent)	3.2	6.9	-10.9	-19.5	9.1	2.9	2.9	3.5	3.2	1.8	2.4	**4.9**	**0.6**
Growth of imports of G&S (US dollar terms, in percent)	11.7	20.9	-8.3	-14.4	4.1	3.8	2.7	2.7	2.7	1.8	1.8	**7.6**	**0.9**
Grant element of new public sector borrowing (in percent)	...	...	...		35.4	35.4	30.7	30.7	30.7	30.7	30.7	**...**	**31.6**
Government revenues (excluding grants, in percent of GDP)	14.0	13.0	10.8	10.2	11.4	12.9	13.4	14.0	14.4	14.9	15.2	**12.9**	**13.6**
Aid flows (in Million of US dollars) 5/	1186.1	-2965.9	-6672.2	253.1	380.0	314.8	359.4	458.0	502.1	534.9	607.9		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	2.9	4.0	3.2	4.2	5.1	5.5	6.1	5.5	**...**	**4.9**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	67.3	92.4	92.3	81.6	81.6	79.9	67.9	72.3	**...**	**77.9**
Nominal GDP (Million of US dollars)	8,409	9,658	8,708	8,601	8,875	9,196	9,531	9,879	10,258	11,088	13,205		
Nominal dollar GDP growth	5.4	14.9	-9.8	-1.2	3.2	3.6	3.6	3.6	3.8	1.8	1.8	**3.1**	**2.2**
Memorandum items:													
PV of external debt 7/	...	...	16.1	17.1	16.2	15.2	14.9	14.9	15.1	21.6	41.1		
In percent of exports	...	...	88.6	115.5	103.5	97.8	96.6	96.4	98.3	145.1	273.3		
Total external debt service-to-exports ratio	5.7	6.1	7.3	9.2	8.9	8.8	8.6	8.4	8.2	11.6	19.8		
PV of PPG external debt (in Million of US dollars)			1406.0	1474.9	1441.5	1401.0	1423.8	1471.2	1548.0	2390.7	5433.7		
(PVt-PVt-1)/GDPt-1 (in percent)				0.8	-0.4	-0.5	0.2	0.5	0.8	2.2	1.2		
Non-interest current account deficit that stabilizes debt ratio	4.0	4.4	-2.7	2.8	2.5	2.2	2.1	2.2	2.4	0.0	2.1		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Ɛα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ɛ=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

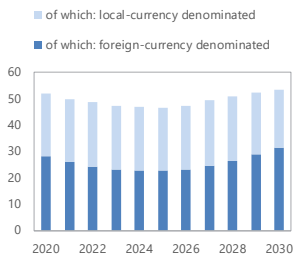
Table 2. Haiti: Public Sector Debt Sustainability Framework, Baseline Scenario, 2017–40

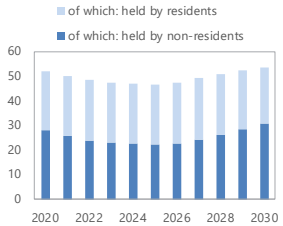
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**38.3**	**39.9**	**47.7**	**51.9**	**49.9**	**48.6**	**47.3**	**46.8**	**46.6**	**53.6**	**70.9**	**35.6**	**49.5**
of which: external debt	**24.2**	**23.5**	**27.4**	**28.0**	**25.8**	**23.8**	**22.9**	**22.5**	**22.4**	**30.8**	**55.8**	**20.1**	**25.2**
Change in public sector debt	-2.5	1.6	7.7	4.3	-2.0	-1.3	-1.2	-0.5	-0.2	1.3	1.7		
Identified debt-creating flows	**-5.4**	**0.5**	**4.8**	**3.0**	**-1.7**	**-1.0**	**-1.0**	**-0.3**	**0.0**	**1.5**	**1.7**	**2.2**	**0.6**
Primary deficit	**0.7**	**2.6**	**3.0**	**6.1**	**3.6**	**3.5**	**3.2**	**3.3**	**3.3**	**2.8**	**2.2**	**3.4**	**3.5**
Revenue and grants	17.7	17.3	12.1	12.4	15.2	15.9	17.2	18.7	19.2	19.7	19.8	**19.1**	**17.9**
of which: grants	3.7	4.3	1.4	2.2	3.8	3.0	3.8	4.6	4.9	4.8	4.6		
Primary (noninterest) expenditure	18.4	19.9	15.2	18.4	18.7	19.4	20.4	22.0	22.5	22.6	21.9	**22.5**	**21.4**
Automatic debt dynamics	**-6.0**	**-2.1**	**1.8**	**-3.0**	**-5.3**	**-4.6**	**-4.1**	**-3.6**	**-3.2**	**-1.3**	**-0.5**		
Contribution from interest rate/growth differential	-2.1	-2.3	-2.1	-2.1	-4.8	-4.2	-3.8	-3.3	-3.0	-1.8	-1.3		
of which: contribution from average real interest rate	-1.6	-1.7	-2.5	-4.1	-4.2	-3.7	-3.2	-2.8	-2.3	-1.1	-0.4		
of which: contribution from real GDP growth	-0.5	-0.6	0.5	2.0	-0.6	-0.5	-0.5	-0.6	-0.6	-0.7	-1.0		
Contribution from real exchange rate depreciation	-4.0	0.2	3.8	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**2.9**	**1.1**	**3.0**	**0.3**	**-0.8**	**-0.7**	**-0.6**	**-0.5**	**-0.5**	**0.3**	**0.9**	**-0.7**	**-0.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**38.2**	**42.5**	**41.3**	**40.8**	**40.0**	**39.7**	**39.7**	**44.7**	**56.9**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**314.5**	**343.7**	**272.5**	**256.5**	**232.4**	**212.7**	**206.4**	**226.4**	**287.6**		
Debt service-to-revenue and grants ratio 3/	**72.6**	**73.1**	**121.7**	**152.0**	**139.5**	**137.0**	**132.2**	**122.3**	**119.8**	**123.3**	**80.6**		
Gross financing need 4/	13.5	15.2	17.8	24.8	24.7	25.3	25.9	26.1	26.3	27.2	18.1		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	1.2	1.5	-1.2	-4.0	1.2	1.0	1.1	1.2	1.4	1.4	1.4	**1.4**	**0.8**
Average nominal interest rate on external debt (in percent)	0.7	0.9	1.0	0.9	0.9	0.9	0.9	1.0	1.1	1.6	1.9	**0.6**	**1.2**
Average real interest rate on domestic debt (in percent)	-11.2	-10.6	-13.4	-17.7	-16.6	-14.5	-12.5	-10.7	-9.1	-4.1	-2.9	**-6.7**	**-9.6**
Real exchange rate depreciation (in percent, + indicates depreciation)	-14.7	0.8	16.3	...	...	...	...	...	...	...	...	**1.3**	**...**
Inflation rate (GDP deflator, in percent)	13.4	12.8	17.3	22.2	21.3	18.3	15.3	12.9	10.9	5.0	5.0	**9.2**	**11.8**
Growth of real primary spending (deflated by GDP deflator, in percent)	-4.8	9.5	-24.6	16.6	3.0	4.7	6.1	9.0	4.0	0.9	1.2	**-0.4**	**4.6**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	3.1	0.9	-4.7	1.8	5.6	4.9	4.4	3.8	3.5	1.5	0.5	**-0.2**	**3.0**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The general government. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Haiti: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020–40

	Projections 1/																				
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
PV of debt-to GDP ratio																					
Baseline	17	16	15	15	15	15	15	16	18	20	22	24	26	28	31	34	36	39	**40**	**41**	**41**
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2020-2030 2/	17	18	19	21	22	23	24	25	27	29	31	33	36	38	**41**	**43**	**45**	**47**	**48**	**48**	**48**
B. Bound Tests																					
B1. Real GDP growth	17	17	17	16	16	17	17	18	20	22	24	26	29	31	34	37	**40**	**43**	**44**	**45**	**45**
B2. Primary balance	17	16	16	15	16	16	16	18	19	21	23	25	27	30	32	35	37	40	**41**	**42**	**42**
B3. Exports	17	18	19	19	19	19	20	21	22	24	26	27	30	32	34	37	39	**42**	**43**	**43**	**43**
B4. Other flows 3/	17	23	28	28	28	28	29	30	31	32	33	35	37	39	**40**	**42**	**44**	**46**	**47**	**47**	**46**
B5. Depreciation	17	21	12	12	12	12	12	13	15	18	21	24	27	31	34	38	**42**	**45**	**48**	**49**	**50**
B6. Combination of B1-B5	17	22	26	26	26	26	27	28	29	30	32	34	35	38	40	**42**	**44**	**46**	**47**	**47**	**47**
C. Tailored Tests																					
C1. Combined contingent liabilities	17	17	16	16	16	16	17	18	20	22	24	26	28	31	33	36	39	**41**	**43**	**43**	**44**
C2. Natural disaster	17	18	18	19	19	20	21	23	25	27	29	32	34	37	**40**	**43**	**46**	**49**	**51**	**51**	**52**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio																					
Baseline	116	104	98	97	96	98	104	111	121	133	145	159	174	**192**	**209**	**227**	**243**	**258**	**268**	**272**	**273**
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2020-2030 2/	116	116	124	133	141	148	162	171	**182**	**196**	**210**	**224**	**240**	**256**	**273**	**288**	**302**	**315**	**323**	**324**	**322**
B. Bound Tests																					
B1. Real GDP growth	116	104	98	97	96	98	104	111	121	133	145	159	174	**192**	**209**	**227**	**243**	**258**	**268**	**272**	**273**
B2. Primary balance	116	104	100	100	101	104	110	118	128	141	153	167	**182**	**199**	**217**	**234**	**250**	**265**	**275**	**278**	**279**
B3. Exports	116	133	159	158	158	161	170	179	**190**	**205**	**219**	**235**	**253**	**274**	**295**	**316**	**335**	**353**	**364**	**367**	**366**
B4. Other flows 3/	116	144	**182**	**182**	**182**	**185**	**195**	**202**	**209**	**217**	**225**	**235**	**246**	**259**	**273**	**286**	**297**	**307**	**312**	**312**	**307**
B5. Depreciation	116	104	61	59	59	60	64	70	81	95	108	124	141	161	**180**	**200**	**218**	**236**	**248**	**254**	**257**
B6. Combination of B1-B5	116	151	160	**183**	**183**	**186**	**196**	**204**	**212**	**223**	**234**	**246**	**260**	**276**	**293**	**309**	**323**	**337**	**344**	**344**	**341**
C. Tailored Tests																					
C1. Combined contingent liabilities	116	106	102	103	104	107	114	122	133	146	158	173	**189**	**206**	**225**	**242**	**259**	**275**	**285**	**289**	**290**
C2. Natural disaster	116	116	118	123	128	134	145	156	170	**186**	**201**	**217**	**235**	**256**	**276**	**296**	**314**	**332**	**344**	**350**	**352**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio																					
Baseline	9	9	9	9	8	8	10	10	10	11	12	11	12	13	15	16	17	18	19	19	20
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2020-2030 2/	9	9	9	9	9	9	11	12	13	14	15	15	16	18	19	21	22	23	23	24	24
B. Bound Tests																					
B1. Real GDP growth	9	9	9	9	8	8	10	10	10	11	12	11	12	13	15	16	17	18	19	19	20
B2. Primary balance	9	9	9	9	8	8	10	10	11	11	12	12	13	14	15	17	18	19	19	20	21
B3. Exports	9	10	12	12	12	11	14	15	16	17	18	17	19	20	22	24	25	26	27	28	28
B4. Other flows 3/	9	9	10	11	10	10	12	15	18	19	20	19	20	21	22	24	25	25	26	26	26
B5. Depreciation	9	9	9	8	7	7	9	9	6	7	8	8	9	10	11	13	14	15	16	16	17
B6. Combination of B1-B5	9	10	11	12	11	11	13	16	19	19	20	19	20	22	23	25	26	27	27	28	28
C. Tailored Tests																					
C1. Combined contingent liabilities	9	9	9	9	9	8	10	10	10	11	12	11	12	14	15	16	18	18	19	20	20
C2. Natural disaster	9	9	9	9	9	9	11	11	11	12	13	13	14	15	16	18	19	20	21	21	22
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio																					
Baseline	13	12	11	10	9	9	10	10	10	11	12	11	12	13	14	16	17	18	**19**	**19**	**20**
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2020-2030 2/	13	12	11	11	10	10	12	12	13	14	15	15	16	17	**19**	**20**	**22**	**22**	**23**	**24**	**24**
B. Bound Tests																					
B1. Real GDP growth	13	13	12	11	10	10	11	11	11	12	13	12	13	14	16	18	**19**	**20**	**20**	**21**	**22**
B2. Primary balance	13	12	11	10	9	9	10	10	11	11	12	12	13	14	15	17	18	**19**	**19**	**20**	**20**
B3. Exports	13	12	11	11	10	10	11	12	13	13	14	14	15	16	17	**18**	**20**	**20**	**21**	**21**	**22**
B4. Other flows 3/	13	12	12	12	11	11	12	15	**18**	**19**	**19**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**25**	**26**	**26**
B5. Depreciation	13	16	14	11	11	10	12	12	8	9	10	10	11	12	14	16	18	**19**	**20**	**21**	**21**
B6. Combination of B1-B5	13	13	12	12	11	11	12	15	17	18	**18**	18	**19**	**20**	**21**	**22**	**24**	**24**	**25**	**25**	**26**
C. Tailored Tests																					
C1. Combined contingent liabilities	13	12	11	10	9	9	10	10	10	11	12	11	12	13	15	16	17	**18**	**19**	**19**	**20**
C2. Natural disaster	13	12	11	10	10	9	11	11	11	12	13	12	13	14	15	17	**18**	**19**	**20**	**20**	**21**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Haiti: Sensitivity Analysis for Key Indicators of Public Debt, 2020–40

	Projections 1/																				
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
PV of Debt-to-GDP Ratio																					
Baseline	42	41	41	40	40	40	40	42	43	44	45	46	46	47	48	49	50	51	53	55	**57**
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2020-2030 2/	42	45	48	49	50	50	49	49	48	47	46	45	45	45	44	44	44	45	45	47	48
B. Bound Tests																					
B1. Real GDP growth	42	44	46	47	47	48	50	52	54	**56**	**57**	**59**	**60**	**62**	**63**	**65**	**66**	**68**	**71**	**73**	**76**
B2. Primary balance	42	44	45	44	43	43	43	45	46	46	47	47	48	49	50	51	51	53	54	**56**	**58**
B3. Exports	42	43	45	44	44	44	44	46	47	48	48	49	50	50	51	52	53	54	**55**	**57**	**59**
B4. Other flows 3/	42	48	55	54	53	53	54	**56**	**57**	**57**	**57**	**57**	**57**	**58**	**58**	**58**	**58**	**59**	**60**	**61**	**62**
B5. Depreciation	42	42	41	39	38	37	37	38	38	38	37	38	37	38	37	38	38	38	39	41	42
B6. Combination of B1-B5	42	42	42	39	39	39	40	41	43	43	44	45	46	47	47	48	49	50	52	54	**56**
C. Tailored Tests																					
C1. Combined contingent liabilities	42	48	47	45	44	44	44	46	47	47	48	48	49	50	51	52	53	54	**56**	**58**	**60**
C2. Natural disaster	42	**66**	**63**	**60**	**58**	**56**	**56**	**57**	**57**	**57**	**58**	**58**	**59**	**60**	**61**	**62**	**63**	**64**	**66**	**68**	**70**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio																					
Baseline	344	273	256	232	213	206	209	217	220	223	226	231	235	239	244	248	253	259	268	278	288
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2020-2030 2/	344	292	290	269	247	238	237	236	232	228	226	225	225	225	226	228	230	234	239	247	254
B. Bound Tests																					
B1. Real GDP growth	344	285	286	265	247	244	251	263	270	276	283	290	298	306	313	321	328	337	349	362	375
B2. Primary balance	344	287	283	254	230	222	223	229	231	233	236	240	244	248	252	256	260	266	274	284	294
B3. Exports	344	283	283	256	235	228	231	238	241	243	245	248	252	255	258	262	265	271	278	287	296
B4. Other flows 3/	344	317	343	312	286	278	281	288	288	288	288	289	290	291	292	293	294	297	302	308	314
B5. Depreciation	344	284	260	230	209	197	193	200	196	196	192	194	191	193	191	194	194	196	202	209	216
B6. Combination of B1-B5	344	276	262	229	210	204	206	214	217	220	223	227	232	236	240	245	249	256	264	274	284
C. Tailored Tests																					
C1. Combined contingent liabilities	344	317	294	263	238	228	229	235	237	239	241	245	249	253	257	261	266	272	281	291	301
C2. Natural disaster	344	436	396	348	309	291	287	290	289	289	291	293	297	301	305	310	315	322	332	343	354
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio																					
Baseline	152	140	137	132	122	120	122	127	129	128	123	117	113	107	99	92	84	77	72	73	81
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2020-2030 2/	152	151	158	158	148	146	149	150	149	143	134	124	115	106	95	85	76	67	61	61	65
B. Bound Tests																					
B1. Real GDP growth	152	144	150	150	143	143	148	156	160	161	157	153	149	144	137	130	123	116	112	114	123
B2. Primary balance	152	140	149	153	138	132	132	135	136	134	128	122	117	111	103	95	88	80	75	77	84
B3. Exports	152	140	137	133	123	120	122	128	131	130	125	119	115	109	101	93	86	79	74	75	82
B4. Other flows 3/	152	140	138	134	124	121	124	130	135	134	129	123	119	113	105	97	90	82	77	79	86
B5. Depreciation	152	133	131	123	111	114	113	115	123	118	117	107	107	97	94	82	80	72	68	68	75
B6. Combination of B1-B5	152	137	136	131	121	119	121	126	128	127	122	117	112	106	99	91	84	76	72	73	80
C. Tailored Tests																					
C1. Combined contingent liabilities	152	140	173	160	143	137	136	138	138	136	130	123	117	111	103	95	87	79	74	75	82
C2. Natural disaster	152	141	269	234	200	183	175	171	167	161	151	142	135	126	117	108	100	91	86	86	93
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.